Zassenhaus Lemma: or butterfly lemma:

Let H, K be subgroups of a group G. And let H^* and K^* be normal groups of H and K. Then we have

(1) $H^*(H\cap K^*) \trianglelefteq H^*(H\cap K)$.

(2) $K^*(H^*\cap K) \trianglelefteq K^*(H\cap K)$

(3) $$\frac{H^*(H\cap K)}{H^*(H\cap K^*)} \cong \frac{H\cap K}{(H^*\cap K)(H\cap K^*)} \cong \frac{K^*(H\cap K)}{K^*(H^*\cap K)}.$$

pf: (1) (2). By symmetry, it suffices to prove (1) & (3).

first, we show that $H^*(H\cap K)$ is a ~~group~~ subgroup of G.

Indeed, $H^* \trianglelefteq H$ and $H\cap K \leq H$. thus $H^*(H\cap K) \leq H \leq G$.

Next, we observe that $K^* \trianglelefteq K$. implies that $(H\cap K^*) \trianglelefteq (H\cap K)$.

take ~~$a\in A_{i-1}, b\in A_i\cap B_{j-1}$~~. $h^*\in H^*$, $k^*\in H\cap K^*$, $h\in H^*$. $k\in H\cap K$.

then $$hk(h^*k^*)(hk)^{-1} = h\underbrace{(kh^*k^{-1})}_{\in H^*}\underbrace{(kk^*k^{-1})}_{\in K^*\cap H}h^{-1} \in H^*(H\cap K^*)H^* = H^*(H\cap K^*)\ \square.$$

(3). just prove: $$\frac{H^*(H\cap K)}{H^*(H\cap K^*)} \cong \frac{H\cap K}{(H^*\cap K)(H\cap K^*)}$$

Define a homomorphism ϕ: $$H\cap K \longrightarrow H^*(H\cap K) \longrightarrow \frac{H^*(H\cap K)}{H^*(H\cap K^*)}.$$
$$h \longmapsto h \longmapsto hH^*(H\cap K^*)$$

$$\ker\phi = \{h\in H\cap K \mid hH^*(H\cap K^*) = H^*(H\cap K^*)\}$$
$$= (H\cap K)\cap(H^*(H\cap K^*)) = (H^*\cap K)(H\cap K^*).$$

thus by first isomorphism theorem. we deduce.

$$\frac{H\cap K}{(H^*\cap K)(H\cap K^*)} \cong \frac{H^*(H\cap K)}{H^*(H\cap K^*)} \quad \square.$$

Corallary. (Jordan–Hölder Thm). Suppose that $\{1\} = A_0 \triangleleft A_1 \triangleleft \cdots \triangleleft A_m = G$
$\{1\} = B_0 \triangleleft B_1 \triangleleft \cdots \triangleleft B_n = G$

then $m=n$. there is a bijection $\sigma: \{1,\dots,m\} \to \{1,\dots,n\}$. s.t. $A_{\sigma(i)}/A_{\sigma(i)-1} \cong B_i/B_{i-1}$

pf (sketched): $$\frac{A_{i-1}(A_i\cap B_j)}{A_{i-1}(A_i\cap B_{j-1})} \cong \frac{(A_i\cap B_j)B_{j-1}}{(A_{i-1}\cap B_j)B_{j-1}}$$

Some Group isomorphism Thm.

(1) $\phi: G \to H$ surjective homomorphism. then $H \cong G/\ker\phi$

(2) $A \leq G$, $B \trianglelefteq G$. Then $AB \leq G$.; $B \trianglelefteq AB$; $A \cap B \trianglelefteq A$.

Thus $\frac{AB}{B} \cong \frac{A}{A \cap B}$

(3) $H \trianglelefteq G$.; $K \trianglelefteq G$; $H \leq K$; then $\frac{K}{H} \trianglelefteq \frac{G}{H}$ and $\frac{G/H}{K/H} \cong \frac{G}{K}$.

(4) (universal property) $N \trianglelefteq G$. then there is a bijection

$$\{H \mid N \leq H \leq G\} \longleftrightarrow \{H' \mid H' \leq G/N\}$$

preserves: intersections; normality; quotients of subgroups.

Direct products; Group actions; Semi-products.

1. Direct product

Thm 1.1. G group. $H \leq G$. $K \leq G$.

(1) $H, G \trianglelefteq G$.

(2) $H \cap K = \{1\}$.

Then $HK \cong H \times K$ (there "$\times$" can be written as "$\otimes$")

pf: (sketched) Define: $\phi: H \times K \to HK \quad (h,k) \to hk$.

Examples: 1. $m, n \in \mathbb{N}_+$. then $\mathbb{Z}_{mn} \cong \mathbb{Z}_m \times \mathbb{Z}_n$ iff $\gcd(m,n) = 1$.

2. Klein groups: $(K_4) V_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$

3. Fudamental thm of finitely generated Abelian groups:

G: finitely generated; Abelian, then

$$G \cong \mathbb{Z}^r \times \mathbb{Z}_{n_1} \times \cdots \times \mathbb{Z}_{n_s}. \quad r \geq 0, \ n_i \mid n_{i+1}$$

② Semi-direct product

Two definitions: ① N, H groups. let $\phi: H \to \mathrm{Aut}(N)$ be a homomorphism.

then $N \rtimes H := \{(n,h) \mid n \in N, h \in H\}$.

$(n_1, h_1)(n_2, h_2) = (n_1 \cdot \phi_{h_1}(n_2), h_1 h_2)$

Thm ②. G group. $N \trianglelefteq G$. $H \leq G$. suppose $H \cap G = \{1\}$.

Then $NH \leq G$ and $NH \cong N \rtimes H$.

Examples: ① $\mathbb{Z}_n \rtimes \mathbb{Z}_n^\times \cong \left\{ \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} \in M_{2\times 2}(\mathbb{Z}_n) \mid a \in \mathbb{Z}_n^\times \right\}$; ② $\mathbb{Z}_n \rtimes \{\pm 1\} \xrightarrow{\cong} D_{2n}$; ③

2. Group action:

Definition 2.1: G group, X set. A left G-action on X is a map:

$$G \times X \to X \qquad (g, x) \mapsto g \cdot x.$$

satisfying 1) $\forall\, x \in X, \ e \cdot x = x$.

2) $\forall\, g, h \in G, \ x \in X, \quad g \cdot (h \cdot x) = (gh) \cdot x$

Remark. Given a group action of G on X $\Longleftrightarrow$ give a homomorphism of S_X.

Examples: 1. S_n on $X = \{1, 2, \ldots, n\}$.

2. D_{2n} on a regular n-gon.

$$D_{2n} \langle r, s \mid r^n = s^2 = 1, \ srs = r^{-1} \rangle.$$

3. ①left translation action: $G \times G \to G$

$$l_g: (g, x) \mapsto gx.$$

②right ~: $G \times G \to G$

$$(g, x) \to xg^{-1}$$

③ Conjugation action: $G \times G \to G$

$$\mathrm{Ad}_g: (g, x) \mapsto gxg^{-1}$$

By Remark. We conclude that: Every group $\cong$ subgroup of S_G.

(Pf. Consider left translation action).

Def. 2.2. (1). An action is faithful iff. $g\cdot x = x$, $\forall x\in X \Rightarrow g=e$.

(2) An action is trivial iff $g\cdot x = x$, $\forall x\in X$.

(3) An action is transitive iff $\forall x,y\in X$. $\exists g\in G$. s.t. $x = gy$

Def 2.3. G acts on X. $\forall\ x\in X$.

① Stabilizer group at x : $= Stab_G(x) = \{g\in G \mid gx = x\}$.

② Orbit of x : $= Orb_G(x) = \{g\cdot x \mid g\in G\} \subseteq X$

Remark ① $Stab_G(x) < G$.

② $X = \coprod_{\text{orbits } O} O$

③ ~~$y\in Stab_G(x)$~~ $y\in Orb_G(x)$, then $Stab_G(y) = g\ Stab_G(x)\ g^{-1}$

Examples: Consider the conjugation action, of G. the orbits are called conjugacy classe

① $G = GL_n(\mathbb{C})$. the representer of one orbit is a Jordan block.

② $G = S_n$, $n = \lambda_1 + \dots + \lambda_n$. $\lambda_1 \geq \dots \geq \lambda_n \geq 1$. $\longleftrightarrow$ one conjugacy classes.
where we use $\tau\,(a_1\ a_2 \dots a_r)\,\tau^{-1} = (\tau(a_1)\ \tau(a_2) \dots \tau(a_r))$.

③ $G = D_{2n}$, ???

Def 2.4. G group, $H < G$, $S \subseteq G$.

(1) $C_G(S) := \{g\in G \mid \forall s\in S,\ gs = sg\}$ is called the centralizer of S

(2) $Z(G) := \{g\in G \mid \forall h\in G,\ gh = hg\}$ is called the center of G.

(3) $N_G(H) := \{g\in G \mid gHg^{-1} = H\}$. is called the normalizer of H.

Remark. (1) Consider the conjugation action on G.
then $Stab_G(g) = C_G(g)$

(2) Consider the conjugation action on $\mathcal{P}(G)$
then $Stab_G(H) = N_G(H)$.

(3) $H \trianglelefteq N_G(H)$.

Def 2.5. G-equivariant : G acts on X and Y. then ϕ is ~ iff.
for $\forall g\in G, x\in X$. we have $\phi(gx) = g\cdot\phi(x)$

Prop. 2.6. G is a transitive action on X. then there is a $\underline{G\text{-equivariant}}$ ① $\underline{\text{bijection}}$ ②.

$$\phi: G/H \xrightarrow{\cong} X,\quad gH \longmapsto gx$$

Corollary 2.7. G acts on X. then $X \cong \coprod_{\text{orbits } O} O = \coprod G/Stab_G(x)$

(Since $Orb_G(x) \cong G/Stab_G(x)$)

and $N_G(P)$ contains a unique Sylow-p subgroup P

Thm. 2.8. G finite group. (action is conjugation).

(1) $\forall g \in G$. the number of elements in conjugacy class is

$$|Ad_G(g)| = |G| / |C_G(g)| = [G : C_G(g)]$$

(2) $g_1, \dots, g_r$ are representatives of conjugacy classes of G. $g_i \notin Z(G)$.

then $|G| = |Z(G)| + \sum_{i=1}^{r} [G : C_G(g_i)]$

let $G = S_5$. the following are Stablizers.

Examples:

S_5	$1+1+\dots+1$	$S_2 \times Z_3$.	$2+3$
$S_2 \times S_3$	$1+1+1+1+2$	$Z_2^2 \rtimes Z_2$	$1+2+2$
$Z_3 \times S_2$	$1+1+3$	Z_4.	$1+4$
Z_5	5		

Application: a structured proof of Sylow theorem.

Def 2.9. $Inn(G) := \{ Ad_g \in Aut(G) \mid g \in G \}$. , $Out(G) := Aut(G) / Inn(G)$

Example: let $G = GL_n(\mathbb{Q})$. $\Rightarrow \ker(Ad) = Z(GL_n(\mathbb{Q}))$

$Ad : G \to Aut(G)$. by linear algebra $\{ a \cdot I_n \mid a \in \mathbb{Q}^\times \} \cong \mathbb{Q}^\times$

Then $Inn(G) \cong GL_n(\mathbb{Q}) / \mathbb{Q}^\times := PGL_n(\mathbb{Q})$ (the projective general linear group)

Def $\psi : GL_n(\mathbb{Q}) \to GL_n(\mathbb{Q})$ $A \mapsto (A^{-1})^T$

Then we have a theory: $Aut(SL_n(\mathbb{Q})) \cong Aut(PGL_n(\mathbb{Q})) \cong \begin{cases} PGL_n(\mathbb{Q}) \rtimes \{1, \psi\} & n \geq 3. \\ PGL_2(\mathbb{Q}) & n = 2 \end{cases}$

where $SL_n(\mathbb{Q}) := \{ A \in GL_n(\mathbb{Q}) \mid \det A = 1 \}$.

Remark. (Dieudonné - O'Meara Thm). the structure of Automorphism of linear groups.

let K be a field. let $n \geq 3$. $\mathrm{char}(K) = 0$.

then $Aut(SL_n(K)) \cong PGL_n(K) \rtimes (Aut(K) \times \langle \psi \rangle) \cong Aut(PGL_n(K))$.

and α $N_G(P)$ contains a unique Sylow p-subgroup P

A structured proof of Sylow's theorems.

Sylow's theorem: $|G| = p^r \cdot m$, $p \nmid m$

(1) Sylow p-subgroups exists.

(2) All sylow p-subgroups are conjugate;

All p-power subgroups is contained in a Sylow-p group.

(3) Define $Syl_p(G) := \{$ Sylow-p-subgroups in $G\}$. $n_p = |Syl_p(G)|$

then 1.). $n_p \equiv 1 \pmod p$

2.) $n_p \mid m$

proof: (1) Suppose (1) is proved for order $< n$.

Let G be the group of order $n = p^r \cdot m$. $p \nmid m$

1.). $r = 0$. then $\{1\} \trianglelefteq G$.

2). $p \nmid |Z(G)|$ then $|G| = |Z(G)| + \sum_{i=1}^{t} [G : C_G(g_i)]$.

there exists $1 \le j \le t$. such that $p \nmid G/C_G(g_j)$.

thus $v_p(|C_G(g_j)|) = r$, by inductive hopothesis,

there exists H, where $|H| = p^r$. $H \le C_G(g_j) \le G$.

3). $p \mid |Z(G)|$, then $Z(G)$ is a finitely generated abelian group.

$\Rightarrow Z(G) = \mathbb{Z}_{p^{r_1}} \times \dots \times \mathbb{Z}_{p^{r_s}} \times \dots =: \mathbb{Z}_{(\ )}(G)_p \times \dots$

$|Z(G)_p| = p^{r'}$, $(r' \ge 1)$

Consider $G \xrightarrow{\pi} G/Z(G)_p := \overline{G}$ $\quad |\overline{G}| = p^{r-r'} \cdot m < n$

then there exists H, where $|\overline{H}| = p^{r-r'}$. $\overline{H} \le \overline{G}$.

thus $H = \pi^{-1}(\overline{H})$, $|H| = p^r$. $H \le G$ □.

(2) let $P \le G$ is a sylow-p subgroup. $Q \le G$. $|Q| = p^{r'}$.

Consider $Q \curvearrowright G/P = \{gP \mid g \in G\}$.

$q \cdot gP := qgP$.

Then $|G/P| = \sum_{i=1}^{t} |Orbits_i| = \sum_{i=1}^{t} |Q/Stab_i|$ is not divisible by p.

thus there exists $Q' := \{q \in Q \mid gP = q \cdot gP\} = Stab_Q(gP) \le Q$

$v_p(|Q'|) = r'$. thus. $Q' = Q$.

thus $\forall\, q \in Q$, we have $qgP = gP \Rightarrow g^{-1}qg \in P$

$\Rightarrow Q \le gPg^{-1}$ □.

Corollary. 1). All sylow subgroups are conjugate.

2). There's only one Sylow-p-subgroup iff one sylow-p group is normal.

3). $N_G(N_G(P)) = N_G(P)$. and $N_G(P)$ contains a unique Sylow-p subgroup P

1).

(3). by (2) we say that the conjugation G-action on $Syl_p(G)$ is transitive

thus $n_p = |Syl_p(G)| = |G/Stab_G(P)| = |G/N_G(P)| = \frac{p^r \cdot m}{p^r \cdot [N_G(P):P]}$.

where P is a sylow-p subgroup

$$|N_G(P)| = |N_G(P)/P| \cdot |P|$$

thus $n_p \mid m$.

2.). Consider P acting on $Syl_p(G)$ by conjugation. (this action is not transitive!).

$$n_p = \sum_{i=1}^{t} |P/Stab_P(P_i)|$$

1°. $Stab_P(P_i) \neq P$. then $Stab_P(P_i) \leq P$. then $p \mid |P/Stab_P(P_i)|$

2°. $Stab_P(P_i) = P$. then $P \subseteq N_G(P_i) \Rightarrow P = P_i$

then $n_p \equiv 1 \pmod p$ □.

Some easy applications:

the Structure of G which has the order of pq. $p \neq q$. p, q primes.

Thm. $p<q$. $|G| = pq$.

① $q \not\equiv 1 \pmod p$: $G \cong Z_{pq}$.

② $q \equiv 1 \pmod p$: 1. $G \cong Z_{pq}$; 2. $G \cong F_{p,q} := \langle a, b \mid a^q = b^p = 1,\ bab^{-1} = a^k \rangle$

where $k^p \equiv 1 \pmod q$